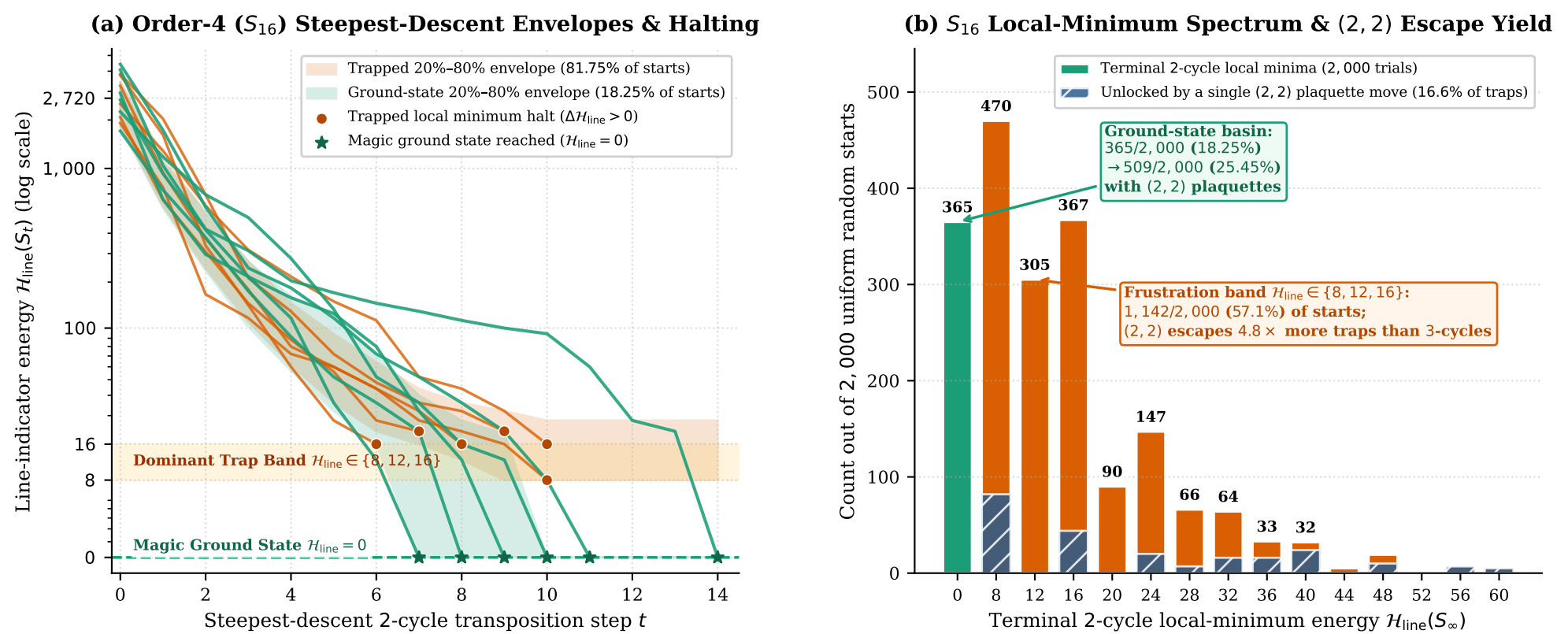

(a) Order-4 (S_{16}) Steepest-Descent Envelopes & Halting
Trapped 20%–80% envelope (81.75% of starts)
Ground-state 20%–80% envelope (18.25% of starts)
Trapped local minimum halt ($\Delta\mathcal{H}_{\text{line}} > 0$)
Magic ground state reached ($\mathcal{H}_{\text{line}} = 0$)
Dominant Trap Band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$
Magic Ground State $\mathcal{H}_{\text{line}} = 0$
Line-indicator energy $\mathcal{H}_{\text{line}}(S_t)$ (log scale)
Steepest-descent 2-cycle transposition step t
(b) S_{16} Local-Minimum Spectrum & $(2, 2)$ Escape Yield
Terminal 2-cycle local minima (2,000 trials)
Unlocked by a single (2, 2) plaquette move (16.6% of traps)
Ground-state basin:
365/2,000 (18.25%)
→ 509/2,000 (25.45%)
with (2, 2) plaquettes
Frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$:
1,142/2,000 (57.1%) of starts;
(2, 2) escapes 4.8× more traps than 3-cycles
Count out of 2,000 uniform random starts
Terminal 2-cycle local-minimum energy $\mathcal{H}_{\text{line}}(S_\infty)$